Jeffrey Mariaraj

rjeffreymariaraj.cs23@rvce.edu.in

linkedin.com/in/jeffrey-mariaraj | github.com/jeffreymariaraj | x.com/ffsj3ff

Bangalore, Karnataka - 560049, India

Objective

Computer Science undergraduate interested in research at the intersection of computational biology, bioinformatics, and AI/ML. Experienced in developing data-driven applications and evaluating machine-learning systems, with particular interests in scientific computing, cheminformatics, efficient AI, and computational approaches to biological and chemical problems. Motivated to contribute to interdisciplinary research through rigorous experimentation, reproducible analysis, and software development.

Experience

- **Deutsche Bank [🌐]** *May 2026 - July 2026*
  Summer Intern Analyst Bangalore, India
  - Developed backend services for an enterprise application using Java and Spring Boot, contributing to maintainable and reliable application functionality.
  - Worked with REST APIs and existing system integrations in a collaborative software engineering environment.
- **Agilisium Consulting – Agilisium Labs [🌐]** *July 2025 - August 2025*
  Project Intern India
  - Developed a sales-forecasting application and REST APIs for data analysis and processing.
  - Built a drug-like molecule generation application as part of the Agilisium Labs internship, using cheminformatics tools for molecular generation, structure validation, and molecular-property analysis.

Education

- **R.V. College of Engineering** *2023-2027*
  Bachelor of Engineering in Computer Science Bangalore, India
  - GPA: 8.51/10
- **Geetanjali Olympiad School** *March 2023*
  High School Bangalore, India
  - Grade: 96.4%

Projects

- **XyloTron: Efficient Local Material Recognition** *2026 – Present*
  Tools: Python, PyTorch, Neural-Network Quantisation, Model Evaluation Ongoing collaboration
  - Investigating quantisation of the XyloTron material-recognition model to support local inference without requiring cloud processing.
  - Evaluating model-compression and deployment trade-offs, with an emphasis on preserving recognition performance while reducing computational and resource requirements.
  - Exploring efficient on-device AI for material recognition, connecting edge computing with practical applications in material identification and sustainable computing.
- **Drug-Like Molecule Generator** *August 2025*
  Tools: Python, RDKit, Streamlit, Pandas
  - Developed an interactive application for generating candidate molecular structures using combinations of chemical fragments and linkers.
  - Used RDKit for molecular structure validation and calculation of properties including molecular weight, LogP, topological polar surface area, and hydrogen-bond donor/acceptor counts.
  - Added molecular visualisation and data export functionality to support inspection and analysis of generated molecular candidates.
- **Intelligent Middleware for AI-IoT Integration** *June 2025*
  Tools: React, Node.js, Python, WebSocket, RESTful APIs, MQTT
  - Designed and implemented a hybrid system combining an API Gateway and MQTT Broker to enable communication between AI agents and IoT devices.
  - Built monitoring, control, and analytics agents alongside device simulators for applications such as smart-home automation and energy management.
  - Measured performance of 1,247 messages/sec, under 100 ms latency, and support for 150 concurrent simulated devices.

- **EchoSQL** *June 2026*
  Tools: Python, FastAPI, React, PostgreSQL, LangChain, LLMs
  - Developed an LLM-powered Text-to-SQL platform that translates natural-language queries into executable SQL.
  - Implemented schema-aware query generation, SQL validation, and execution against PostgreSQL in a full-stack application.

PUBLICATIONS

[1] **Jeffrey Mariaraj**, et al. "An Intelligent Intermediary System for Seamless Integration of AI Agents with MQTT-based IoT Devices."
2025 IEEE International Conference on Computational Systems and Information Technology for Sustainable Solutions (CSITSS 2025). Published in the *IEEE Xplore Digital Library*.
DOI: 10.1109/CSITSS67709.2025.11295492

SKILLS

- **Programming Languages:** Python, JavaScript, C, Java
- **Data Science & Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Deep Learning, CNNs, Computer Vision, Model Evaluation
- **Computational Biology & Cheminformatics:** RDKit, Molecular Structure Validation, Molecular Property Analysis
- **Scientific Computing & Data Analysis:** NumPy, Pandas, SciPy, Simulation, Statistical Analysis, Performance Evaluation
- **Web Technologies & APIs:** React, Node.js, WebSocket, RESTful APIs, FastAPI, Spring Boot, Streamlit
- **Database Systems:** PostgreSQL, MongoDB, Data Processing
- **Cloud, Edge & DevOps:** AWS (EC2, S3), Google Cloud Platform, Azure IoT Hub, Docker, Git, GitHub Actions, Edge Computing
- **Specialized Areas:** Neural-Network Quantisation, Efficient AI, AI-IoT Systems, LLM Integration, Molecular Generation
- **Research Skills:** Literature Review, Simulation and Modelling, Reproducible Experiments, Algorithm Design, Data Analysis, Performance Evaluation